

The Exact Spectral Gap of Kac's Master Equation: A Complete Conditional-Projection Induction

HCS-C322 theorem-and-evidence package

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Abstract

We give a normalization-locked proof of the exact gap for Kac's uniform binary-collision master equation. On $S^{N-1}(\sqrt{N})$, with unordered-pair angle average Q_N and positive generator $L_N = N(I - Q_N)$, the gap is $(N+2)/[2(N-1)]$. We reproduce the infinite-dimensional lower induction via coordinate conditional expectations, identify the quartic equality mode and its multiplicity, and close the $N = 2$, positive-energy, and zero-energy boundaries. Exact polynomial matrices audit the algebra but are not offered as the lower-bound proof.

1 Frozen collision operator

Let σ_N be normalized surface measure on $X_N = S^{N-1}(\sqrt{N})$, $N \geq 2$. If $R_{ij}(\theta)$ rotates the (i, j) coordinate plane, define

$$Q_N f(v) = \binom{N}{2}^{-1} \sum_{i < j} \frac{1}{2\pi} \int_{-\pi}^{\pi} f(R_{ij}(\theta)v) d\theta, \quad L_N = N(I - Q_N). \quad (1)$$

Thus pairs are unordered, the angular measure is a probability, and the factor N belongs to the generator. Kac introduced this collision model in kinetic theory [1]; the exact gap architecture below is due to Carlen, Carvalho, and Loss [2].

Each pair average is the orthogonal projection onto the functions invariant under its circle action. Hence Q_N is a positive self-adjoint Markov contraction. Its fixed functions are constants: coordinate-plane rotations generate $SO(N)$, which is transitive on X_N . Put

$$\lambda_N = \sup_{\substack{\|f\|_2=1 \\ \langle f, 1 \rangle=0}} \langle f, Q_N f \rangle, \quad \Delta_N = N(1 - \lambda_N).$$

Let

$$F_N(v) = \sum_{i=1}^N v_i^4 - \frac{3N^2}{N+2}.$$

The centering uses $\mathbb{E}_{\sigma_N} v_i^4 = 3N/(N+2)$. Direct angle averaging gives

$$\frac{1}{2\pi} \int_{-\pi}^{\pi} \{(u \cos \theta + v \sin \theta)^4 + (-u \sin \theta + v \cos \theta)^4\} d\theta = \frac{3}{4}(u^2 + v^2)^2. \quad (2)$$

Summing (2) over pairs and using $\sum_i v_i^2 = N$ yields

$$Q_N F_N = \left(1 - \frac{N+2}{2N(N-1)}\right) F_N, \quad L_N F_N = \frac{N+2}{2(N-1)} F_N. \quad (3)$$

This is an upper bound for the gap, not yet a proof that no slower mode exists.

Revision certificate: operator normalization and quartic upper bound

This original round freezes the master generator and computes the exact quartic candidate rate without claiming the missing lower bound.

References

- [1] M. Kac, "Foundations of kinetic theory," in *Proc. Third Berkeley Symp. Math. Statist. Probab.*, vol. III (1956), 171–197, UC Berkeley Library record 112857.
- [2] E. A. Carlen, M. C. Carvalho, and M. Loss, "Determination of the spectral gap for Kac's master equation and related stochastic evolution," *Acta Math.* 191 (2003), DOI: 10.1007/BF02392695; arXiv:math-ph/0109003.